

MACRO INTELLIGENCE AGENT — RUNIC AGENT v2.2

Complete Build Specification | May 26, 2026 | Confidential — Internal Use Only

Hi Divyanshu,

This is the complete and final specification for the Macro Intelligence Agent (also called the Runic Agent). Everything you need to build v1 is in this document and the attached files. Read all attachments before writing a single line of code.

Attached files: (1) Combo Cheat Sheet PDF — the complete variable and combo reference; (2) Sample BTIG Note PDF — the exact format the nightly output must match; (3) Methodology Note PDF — threshold justification; (4) SSI Open Questions DOCX — tests to run; (5) Macro_Intelligence_Agent_Spec.xlsx — original variable list.

SECTION 1 — WHAT WE ARE BUILDING

A Python agent that runs nightly alongside the C++ model. It reads 12 macro variables, checks 298 variable combinations for simultaneous extremes, classifies the current macro regime via Claude API, finds the closest historical analogs, calculates regime-adjusted hit rates, and generates a nightly briefing note. It writes one JSON file that the C++ model reads every 5 minutes.

The combo count is 298: 12C1 = 12 single-variable signals. 12C2 = 66 two-variable combos. 12C3 = 220 three-variable combos. Of these, 7 are pre-named (A through G) with validated history. The remaining 291 are auto-discovered at runtime.

SECTION 2 — THE 12 VARIABLES AND FRIDAY DATA PULL

Run the following as a scheduled job every Friday. Use percentile rank (3-year rolling window) — NOT z-scores. Financial distributions have fat tails; z-scores understate extremes. Percentile rank is distribution-agnostic.

#	Variable	Source / ticker	Compute	RARE threshold	EXTREME threshold	Combo
1	NFCI	FRED: NFCI	Weekly value. 3yr pctile rank.	+0.3 SD $\approx$ 80th pctile	+0.8 SD $\approx$ 95th pctile	A, E
2	HY Credit Spreads OAS	FRED: BAMLH0A0HYM2	Daily OAS in bps. 3yr pctile.	>400bps OR >+1.0 SD	>500bps OR >+2.0 SD	A, B, F, G
3	Fed Balance Sheet MoM $\Delta\%$	FRED: WALCL	MoM % change.	$\pm 0.8\%$ MoM	$\pm 2.0\%$ MoM	A, C
4	USD/CNH 4wk % change	Yahoo: USDCNH=X	4-week % change.	$\pm 1.5\%$	$\pm 3.5\%$	A, C, G
5	WTI Oil 4wk % change	Yahoo: CL=F	4-week % change.	$\pm 6\%$	$\pm 10\%$	C
6	VIX Level	CBOE via Yahoo: ^VIX	Daily close. 3yr pctile.	>25 AND >80th pctile	>35 AND >95th pctile	B, D, G
7	VIX Term Structure VIX3M $\div$ VIX	Yahoo: ^VIX3M / ^VIX	Compute ratio daily.	<0.95 or >1.10	<0.85 or >1.20	D, G
8	CFTC Net Spec (TFF report)	CFTC.gov — Lev_Money_Positions AND Asset_Mgr_Positions	Both columns, 3yr pctile rank separately.	<15th or >85th pctile	<5th or >95th pctile	B, D, E, F

9	10Y-2Y Yield Curve	FRED: T10Y2Y	Spread in bps + 4wk change.	<-30bps OR steepen >15bps/4wk	<-80bps OR steepen >40bps/4wk	A, E
10	CPI Surprise	BLS.gov vs Investing.com calendar	Actual minus consensus.	±0.2pp	±0.4pp or 2× consec hot	C
11	Gold/Silver Ratio (GSR)	Yahoo: GC=F / SI=F	4-week % change of ratio.	4wk Δ > ±5%	4wk Δ > ±8%	A+
12	CAPE / Shiller P/E	multpl.com scrape (monthly)	Absolute level.	>28x or <16x	>32x or <12x	E

SECTION 3 — SIGNAL ENGINE LOGIC

3.1 Combo Detection

For each of the 298 combinations: compute rolling 3-year percentile rank for each variable. Check if ALL required variables are simultaneously at or beyond the RARE threshold. If yes: record a combo fire in the `combo_fires` table with the date, variable IDs, combo type, and named combo (A–G or null for unnamed combos).

3.2 Named Combo Special Rules

- Combo B: requires VIX >25 AND HY >400bps AND CFTC <15th pctile ALL simultaneously. A single variable or two firing is a WATCH flag, not Combo B.
- Combo C duration tracker: once WTI 4wk Δ >+10%, start a week counter. Bucket: SHORT <6 weeks, MEDIUM 6–16 weeks, LONG >16 weeks. Store in `combo_fires.duration_bucket`.
- Combo C cancel condition: WTI 4wk Δ stays below +5% for 4 consecutive weeks AND no hot CPI surprise in the same 4-week window. BOTH legs must clear. Cancel = remove ACTIVE flag.
- Combo F: active for 26 weeks from fire date. No early cancel unless Combo B fires again.

3.3 CRITICAL — VIX Multiplier Bypass Rule

THIS RULE PREVENTS THE DALIO OCTOBER 2022 ERROR. Add this as a named constant and docstring in the code:

```
# CRITICAL RULE: VIX_REGIME_BYPASS # When runic_combo = 'B' (Maximum Capitulation) is confirmed active, # the VIX regime size multiplier is BYPASSED. Do NOT reduce position # size during a confirmed Combo B. Historical basis: Oct 13, 2022 — # VIX 33.6 would have triggered 0.75x multiplier but Combo B fired # simultaneously. Multiplier would have cut size 25% at the exact # market bottom. Combo B overrides the multiplier. # Same bypass applies to Combo F when SSI confirms (>=2 of 4 signals).
```

SECTION 4 — TWO VALIDATION EXAMPLES

Build unit tests that reproduce these two exact states using historical FRED and Yahoo Finance data. If the engine does not fire on these exact dates, fix the signal engine before proceeding.

Example	Date	What the inputs looked like	Expected output	Actual outcome
Example 1 Combo B	Oct 13, 2022	VIX: 33.6 (>25 ✓). HY OAS: 614bps (>400 ✓). CFTC Fast Money: ~8th pctile 3yr (<15th ✓). All 3 conditions met.	Combo B fires. Regime: {fed_cycle: HIKING_LATE, curve_regime: INVERTED, geo_overlay: SANCTIONS, val_regime: ELEVATED, liquidity: GLOBAL_TIGHT}. Hit rate regime-adjusted: 68%. VIX multiplier BYPASSED (Combo B active). Recommendation: BULLISH contrarian, full size.	SPX +19% over next 3 months. System correct.

Example 2 Combo F	Jun 29, 2020	SPX reclaimed 50-Week MA week of Jun 26, 2020 with +5.1% weekly gain (>3% ✓). CFTC Fast Money: 31st pctile (<50th ✓ — still net short post-COVID).	Combo F fires. Regime: {fed_cycle: QE, curve_regime: NORMAL, geo_overlay: PANDEMIC, val_regime: ELEVATED, liquidity: GLOBAL_EASY}. Hit rate in QE/easy regime: 91%. Full size. Recommendation: BULLISH.	SPX +18.3% over next 6 months. System correct.
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SECTION 5 — CLAUDE API INTEGRATION

5.1 Setup

- API endpoint: <https://api.anthropic.com/v1/messages>
- Model string: claude-sonnet-4-6 (always use this — do not substitute older models)
- Set environment variable: ANTHROPIC_API_KEY
- If model string returns 404: call GET <https://api.anthropic.com/v1/models> with the API key to list available models and find the current Sonnet string.
- pip install anthropic — cleaner than raw requests
- JSON parsing: always strip markdown fences before json.loads(). Use: `text.replace('```json', '').replace('```', '').strip()`

5.2 Place 1 — Macro Regime Classifier (runs once per combo fire date)

Purpose: classify each combo fire date into 5 regime dimensions. Cost: <\$0.001 per call. Full historical backfill of 400 dates = <\$0.50.

system: "You are a macro regime classifier. Return ONLY valid JSON, no preamble, no explanation, no markdown." user: "Date: {DATE}. Fed funds rate: {FFR}. WALCL 8-week direction: {WALCL_DIR}. 10Y-2Y spread: {CURVE}bps, direction: {CURVE_DIR}. NFCI sign: {NFCI_SIGN}. CAPE decile: {CAPE_DECILE}. Classify: {fed_cycle: HIKING_EARLY|HIKING_LATE|CUTTING_EARLY|CUTTING_LATE|PAUSING|QE|QT, curve_regime: INVERTED|FLAT|STEEPENING|NORMAL, geo_overlay: NEUTRAL|TRADE_WAR|SANCTIONS|REGIONAL_WAR|PANDEMIC|FINANCIAL_CRISIS, val_regime: EXTREME|ELEVATED|FAIR|CHEAP, liquidity: GLOBAL_EASY|GLOBAL_TIGHT}" max_tokens: 150

5.3 Place 2 — Nightly Briefing Note

Purpose: generate the nightly macro intelligence note. Output format must match the attached sample PDF exactly. See Section 6 for output structure.

system: "You are a senior macro strategist. Write with the structure and precision shown in the attached sample note: (1) one sentence dominant signal statement with the reason it outweighs competing signals; (2) why it dominates, with specific variable levels and dates; (3) the two closest historical analogs with exact dates and forward returns; (4) one sentence action recommendation. 200–250 words. No hedging language." user: "Active combos: {COMBO_LIST with duration}. Regime: {REGIME_JSON}. Analog dates: {ANALOG_LIST with SPX returns}. All 12 variable readings: {VAR_JSON}. Dominant signal: {DOMINANT_COMBO}. Direction: {BRAVE_FEARFUL}." max_tokens: 500

SECTION 6 — JSON OUTPUT FORMAT

Written nightly to shared directory. C++ model polls every 5 minutes.

```
{ "date": "2026-05-26", "regime": { "fed_cycle": "CUTTING_EARLY", "curve_regime": "STEEPENING", "geo_overlay": "REGIONAL_WAR",
"val_regime": "EXTREME", "liquidity": "GLOBAL_EASY" }, "dominant_signal": "C", "dominant_reason": "WTI shock week 11 of 16 (MEDIUM
duration). Bearish 83% hit rate at 3m horizon overrides bullish Combo F at tactical timeframe.", "brave_fearful":
"TACTICAL_FEARFUL_STRATEGIC_BRAVE", "active_combos":
[{"combo": "C", "duration_weeks": 11, "duration_bucket": "MEDIUM"}, {"combo": "E", "status": "PARTIAL"}], "watch_combos": ["D"], "ssi_multiplier": 1.0,
"analog_dates": ["2008-06-15", "2022-06-16"], "spx_3m_forward_avg": -3.2, "spx_3m_hit_rate": 0.83, "combo_f_active": true,
"combo_f_weeks_elapsed": 8, "narrative": "[Claude API generated text — 200-250 words matching sample PDF format]" }
```

SECTION 7 — DATABASE

7.1 Use SQLite for v1

SQLite = single .db file, zero server setup, perfect for one writer (nightly Python) and one reader (C++ model). Do not over-engineer for v1. The schema is designed so that migration to PostgreSQL (AWS RDS) for the live web app requires only changing the connection string — no code changes needed.

7.2 Key Tables

Table	Purpose	Key fields
variables	12 variables master list	var_id PK, name, source_ticker, paradigm, speed, combo_slots
thresholds	RARE + EXTREME per variable per direction	threshold_id PK, var_id FK, tier, direction, value, value_type
daily_readings	Time series all 12 variables	date, var_id FK, raw_value, pctile_rank_3yr, signal_tier
signal_fires	When a variable crossed into RARE/EXTREME	fire_id PK, var_id FK, date, tier, direction, weeks_in_tier
combo_fires	298-combo engine output	combo_id PK, date, var1_id, var2_id, var3_id, runic_combo, macro_regime JSON, duration_bucket
forward_returns	SPX/NDX returns from every combo fire	return_id PK, combo_id FK, spx_1w, spx_2w, spx_1m, spx_3m, spx_6m
rule_library	Named rules with hit rates	rule_id PK, rule_name, signal_condition, n_obs, hit_rate, avg_return_3m, regime_hit_rates JSON

7.3 Hit Rate SQL (include this in the codebase as a named function)

Raw hit rate for Combo B (bullish — SPX higher 3m later) SELECT CAST(SUM(CASE WHEN fr.spx_3m > 0 THEN 1 ELSE 0 END) AS FLOAT) / COUNT(*) FROM combo_fires cf JOIN forward_returns fr ON cf.combo_id = fr.return_id WHERE cf.runic_combo = 'B' # Regime-adjusted (Fed cutting cycles only) -- add: AND json_extract(cf.macro_regime,'\$.fed_cycle') LIKE 'CUT%'

SECTION 8 — BUILD SEQUENCE AND TIMELINE

Phase	Task	Days	Notes
1	Data layer: all 12 Friday pulls as scheduled jobs. SQLite schema. Historical backfill 1990–2026.	2–3	Test each data source individually. Verify CFTC TFF column names exactly.
2	Signal engine: percentile rank engine, 298-combo detector, named combo special rules (C duration, F window, B bypass).	4–5	Build unit tests for Oct 2022 and Jun 2020 first. Engine must pass both before moving on.
3	Claude API: regime classifier (batch backfill all historical fire dates), nightly narrative generator.	1–2	Test regime classifier on 10 known dates manually. Verify JSON parsing strips markdown fences.
4	Forward returns table: for each historical combo fire, pull SPX returns at 1w/2w/1m/3m/6m.	2–3	Use yfinance for SPX (^GSPC). Handle weekends and holidays correctly.
5	JSON output + C++ integration + nightly scheduler.	1–2	Cron job: 0 21 * * 1-5. Test C++ reads new JSON correctly.
TOTAL		10–15 days	Do not skip the unit tests in Phase 2.

Questions? Call before writing code — it is faster than re-reading and re-doing.

Best, Rohit